Omega-3 + RT on metabolic syndrome

Outline

- Primary (logistic regression)
  - Does omega-3 supplementation, strength training, or their combination reduce the prevalence of metabolic syndrome compared to placebo?
- Secondary (LMM)
  - How do the interventions affect each component of metabolic syndrome?
  - Are the effects additive or synergistic?
- Exploratory
  - Does supplementation compliance, verified by RBC EPA+DHA percentage, predict improvements in metabolic outcomes?
  - Do baseline characteristics (sex, age, BMI, initial metabolic syndrome components) influence responsiveness?
- For discussion:
  - Feasibility, effectiveness, and clinically meaningful changes?

Methods

Participants

Table 1: Participant characteristics.

	N-3(♂ n = 32; ♀ n = 42)	Placebo(♂ n = 35; ♀ n = 38)
	Male	
Age (years)	47 (8) [29 - 60]	49 (6) [36 - 59]
Stature (cm)	180 (8) [153 - 193]	181 (6) [165 - 192]
Body mass (kg)	96 (17) [69 - 152]	98 (18) [71 - 142]
Body mass index ($kg \times m^{-2}$)	29 (5) [20 - 45]	30 (6) [23 - 52]
	Female	
Age (years)	45 (8) [30 - 59]	47 (9) [23 - 60]
Stature (cm)	169 (6) [158 - 182]	168 (6) [152 - 180]
Body mass (kg)	83 (17) [58 - 136]	79 (14) [53 - 114]
Body mass index ($kg \times m^{-2}$)	29 (6) [18 - 45]	28 (5) [20 - 40]

Data are presnted as Mean (SD) [Min-Max]

Statistics and data anlysis

Results

Missing data analysis

To assess if our outcome variables predict drop-out we fit a logistic regression model

$$
\begin{aligned}
D_i &\sim \text{Bernoulli } (p_i) \\
\text{Logit } (p_i) &= \alpha_{[A]} + \beta_{X[A]} X_i + \beta_{S[A]} S_i + \beta_{T[A]} T_i + \beta_{_{TX[A]}} T_i X_i \\
\alpha_{[A]}, \beta_{X[A]}, \beta_{S[A]}, \beta_{T[A]}, \beta_{TX[A]} &\sim \text{Normal } (0, 0.5)
\end{aligned}
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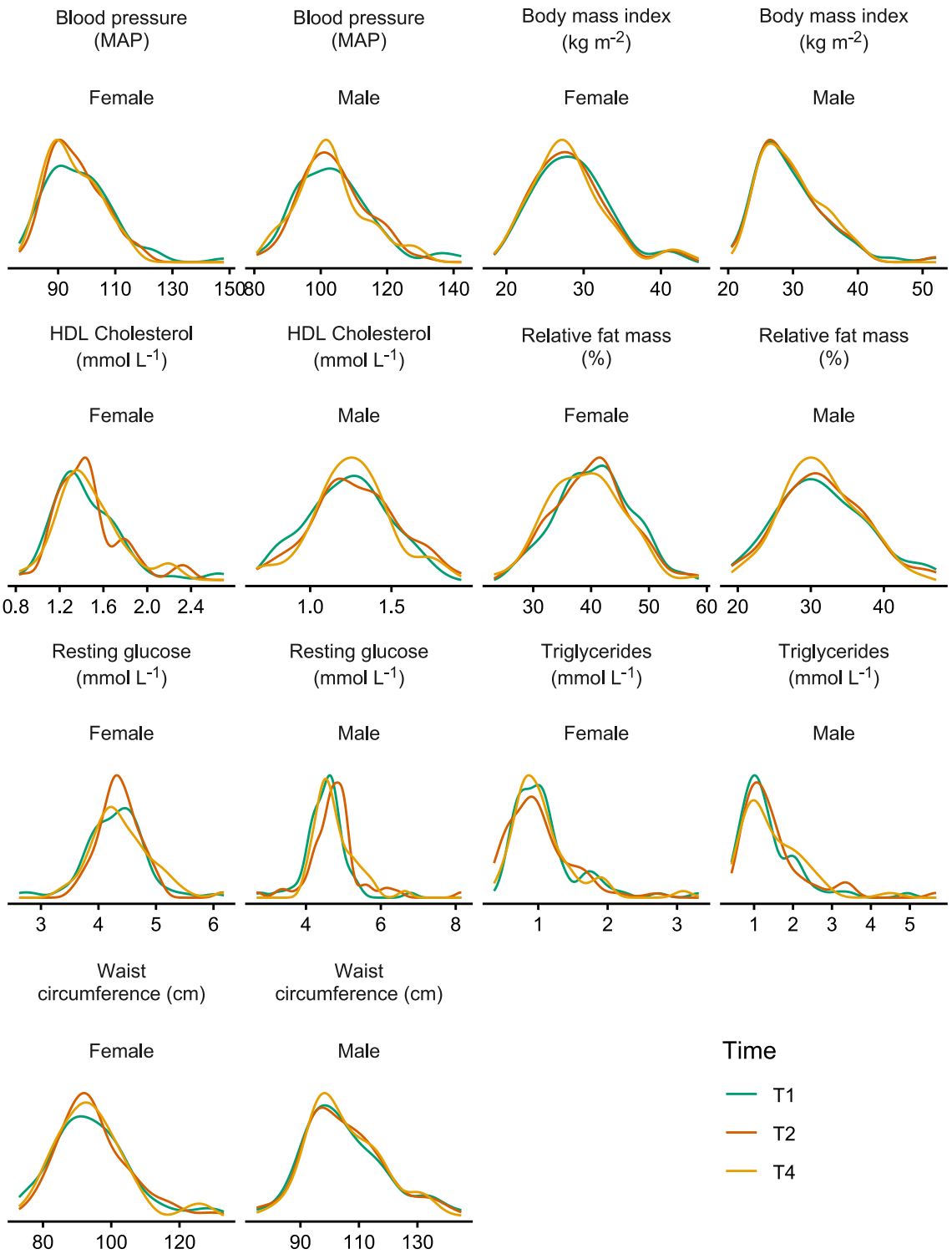


Figure 1: Observed data distributions of main outcomes.

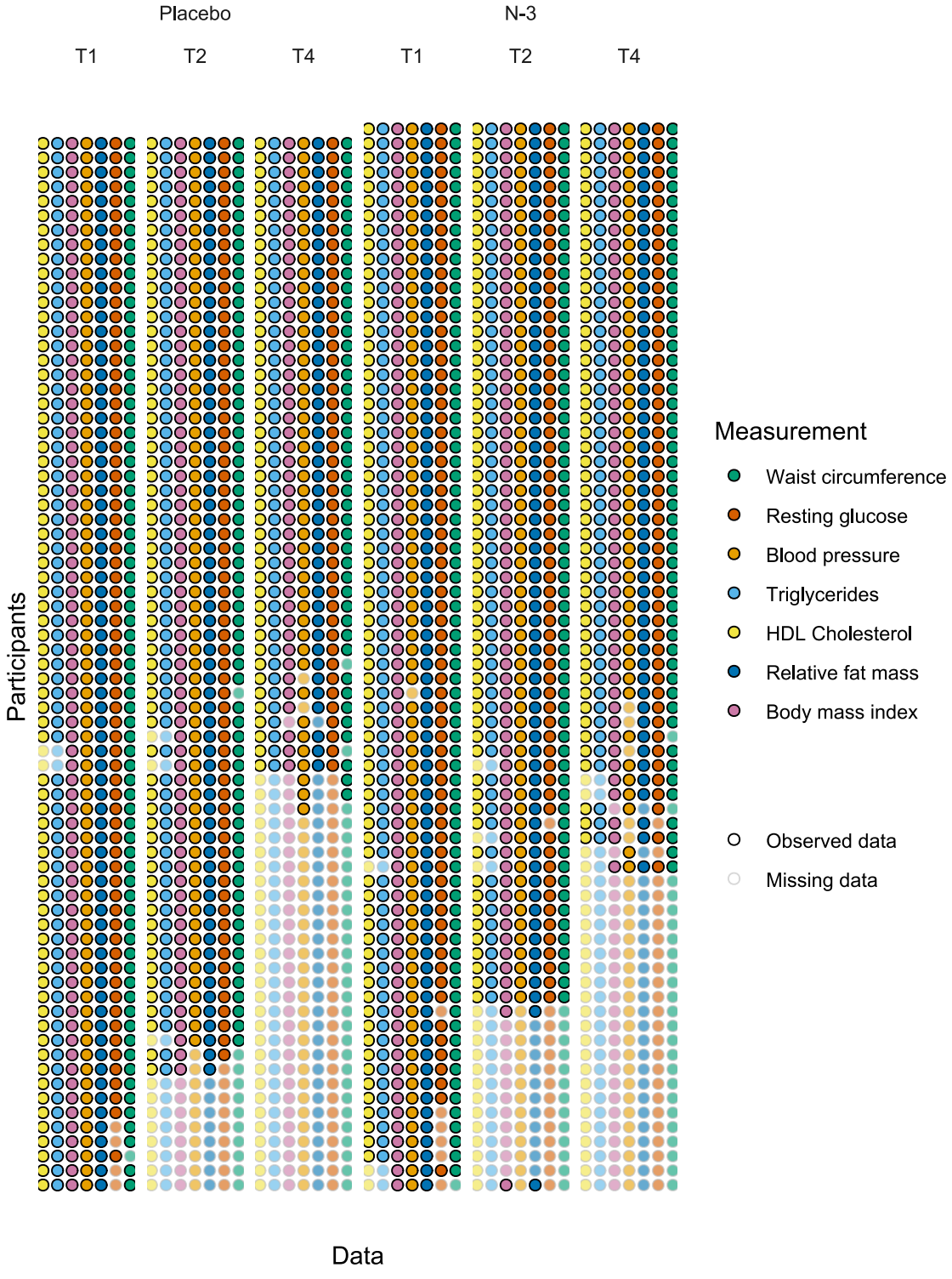


Figure 2: Observed and missing data across treatment groups and time-points. Rows are participants, columns are data points.

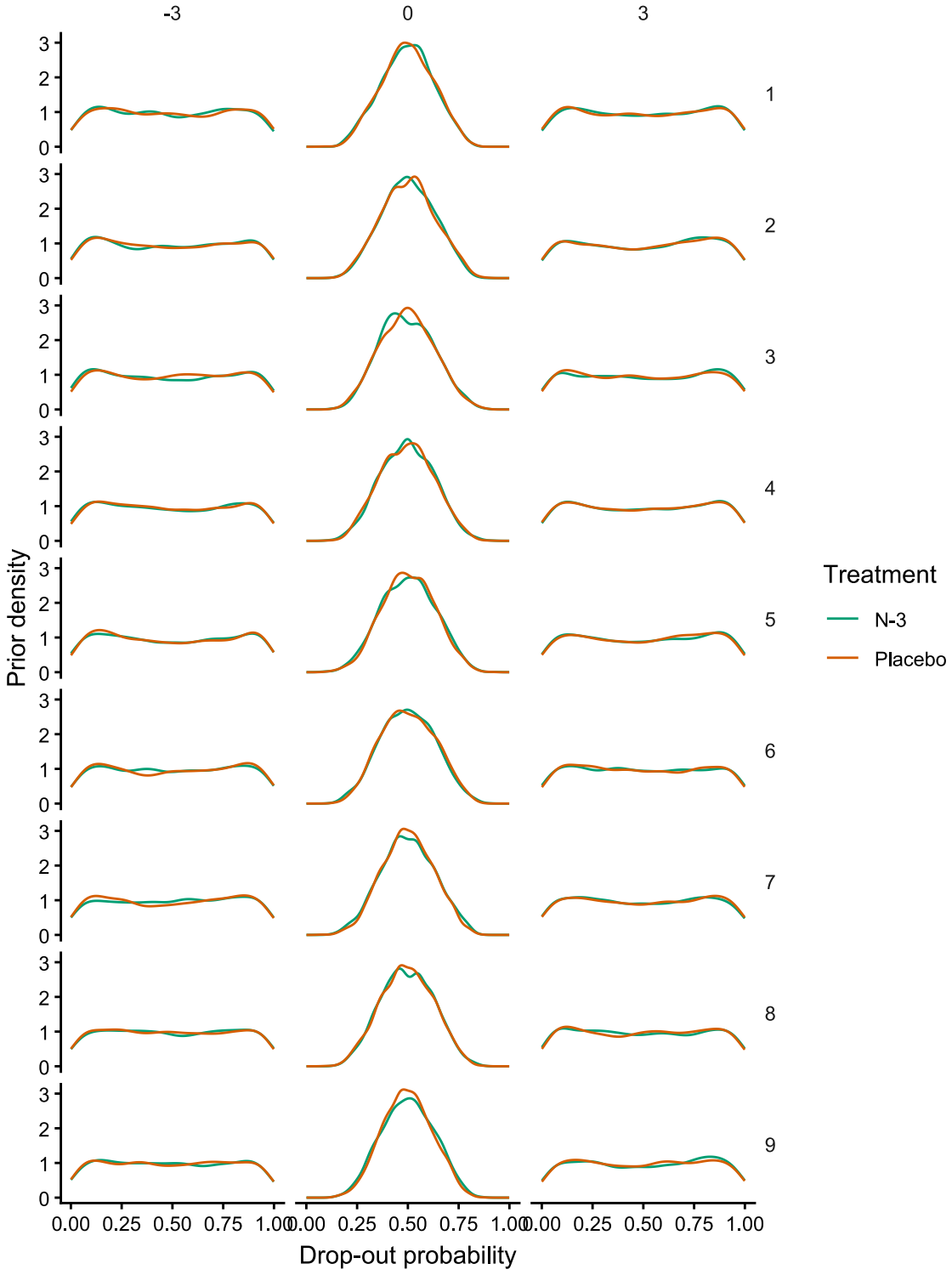


Figure 3: Prior predictive simulation for model 1. Columns are values of the predictor (SD-units) rows represents each predictor.

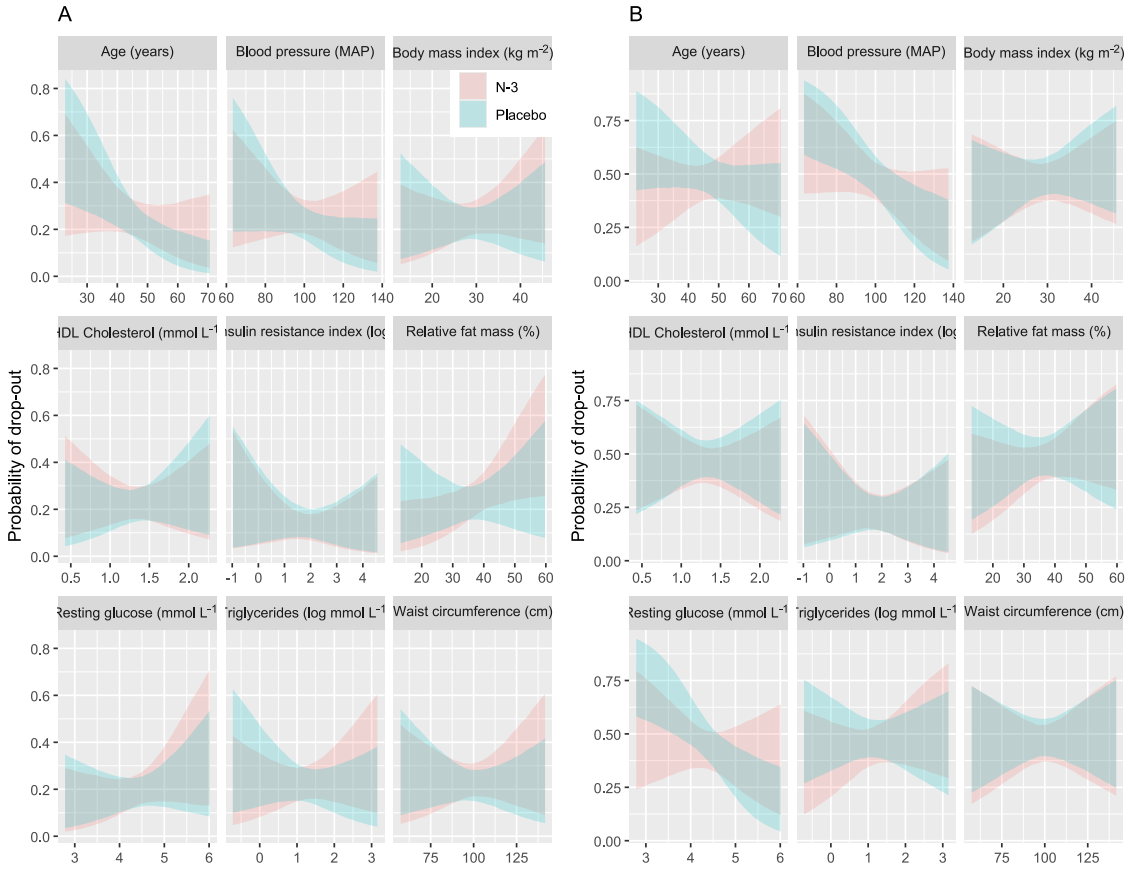


Figure 4: Probability of drop-out across values of baseline co-variates on per treatment group to time-point 2 (A) and 4 (B). Estimates are adjusted for sex, intervals are 90% percentile interval.

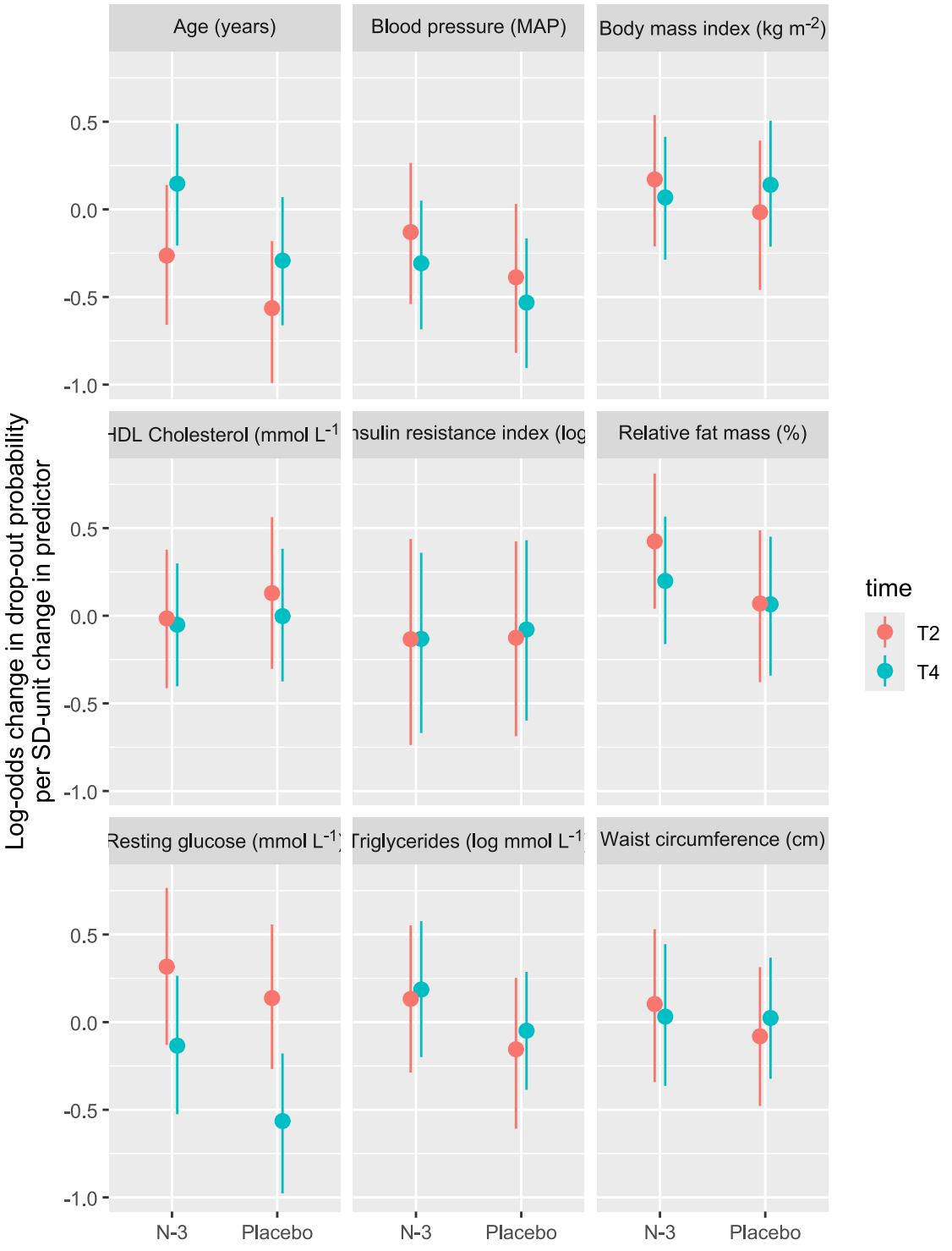


Figure 5: Effects of one SD-uniut change in baseline value on log-odds probability of droup-out per treatment group. Estimates are adjusted for sex, intervals are 90% percentile interval.

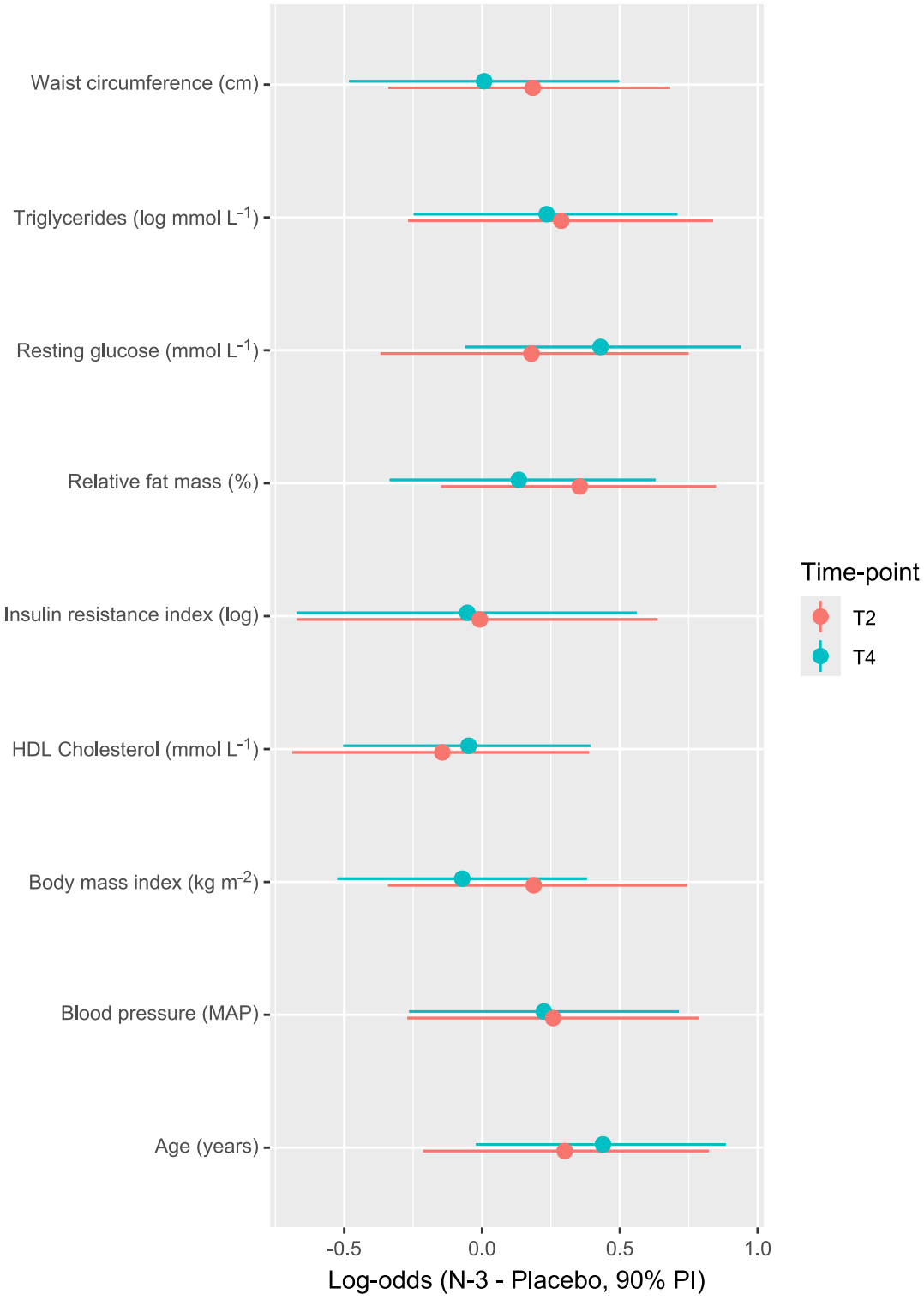


Figure 6: Differences in the effect of baseline value on drop-out probability between treatments groups. Estimates are adjusted for sex, intervals are 90% percentile interval.

N-3 supplementation and fatty acid profiles

Fatty acids, including including EPA (C205n3) and DHA (226n3) where measured in whole blood. We will assess the effect of supplementation on group level fatty acid profiles using mixed-effects models. As preliminary models indicated heteroscedasticity of residuals, log-transformed data was used in modelling.

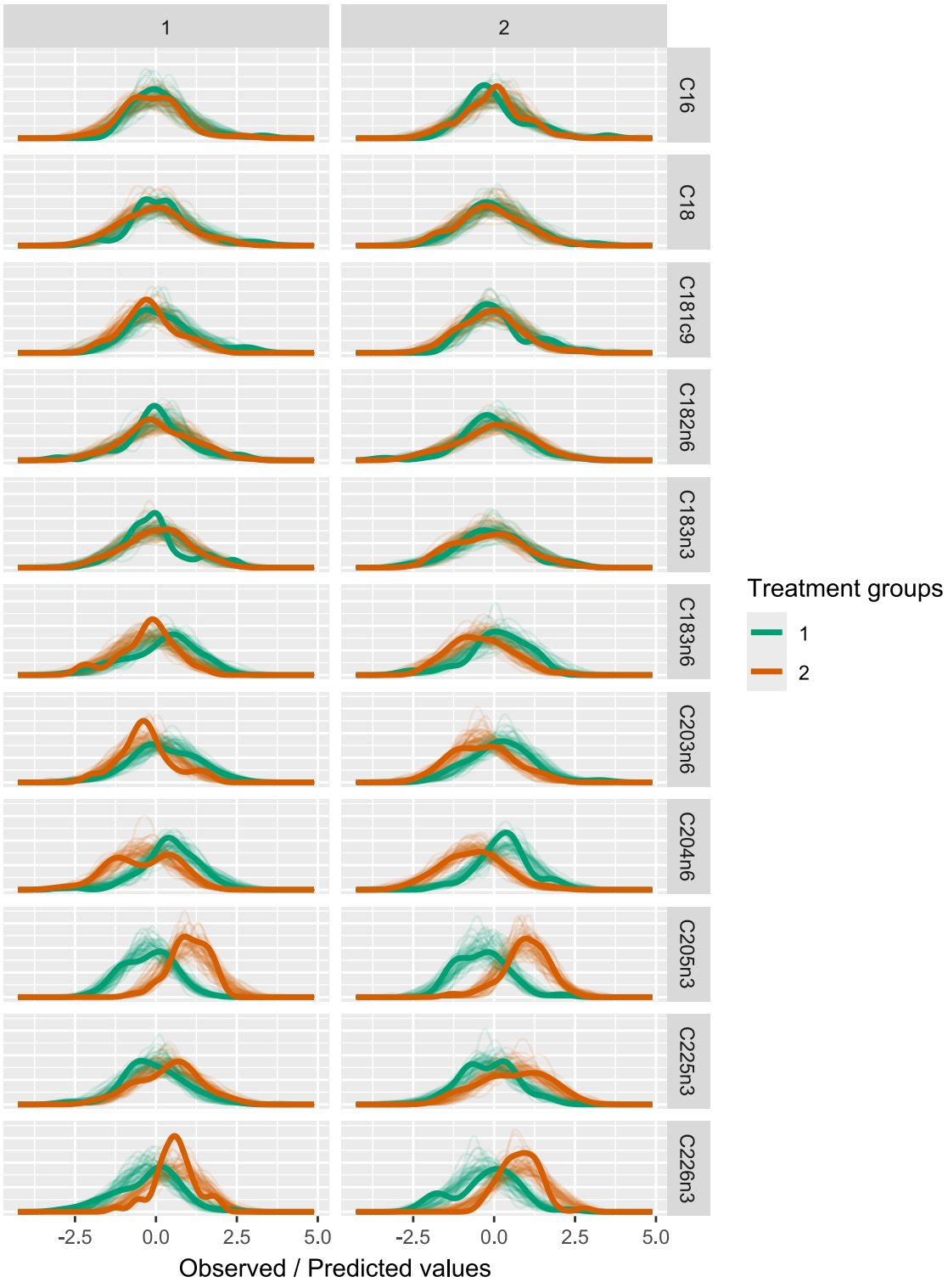


Figure 7: Posterior predictive check of Model 2.

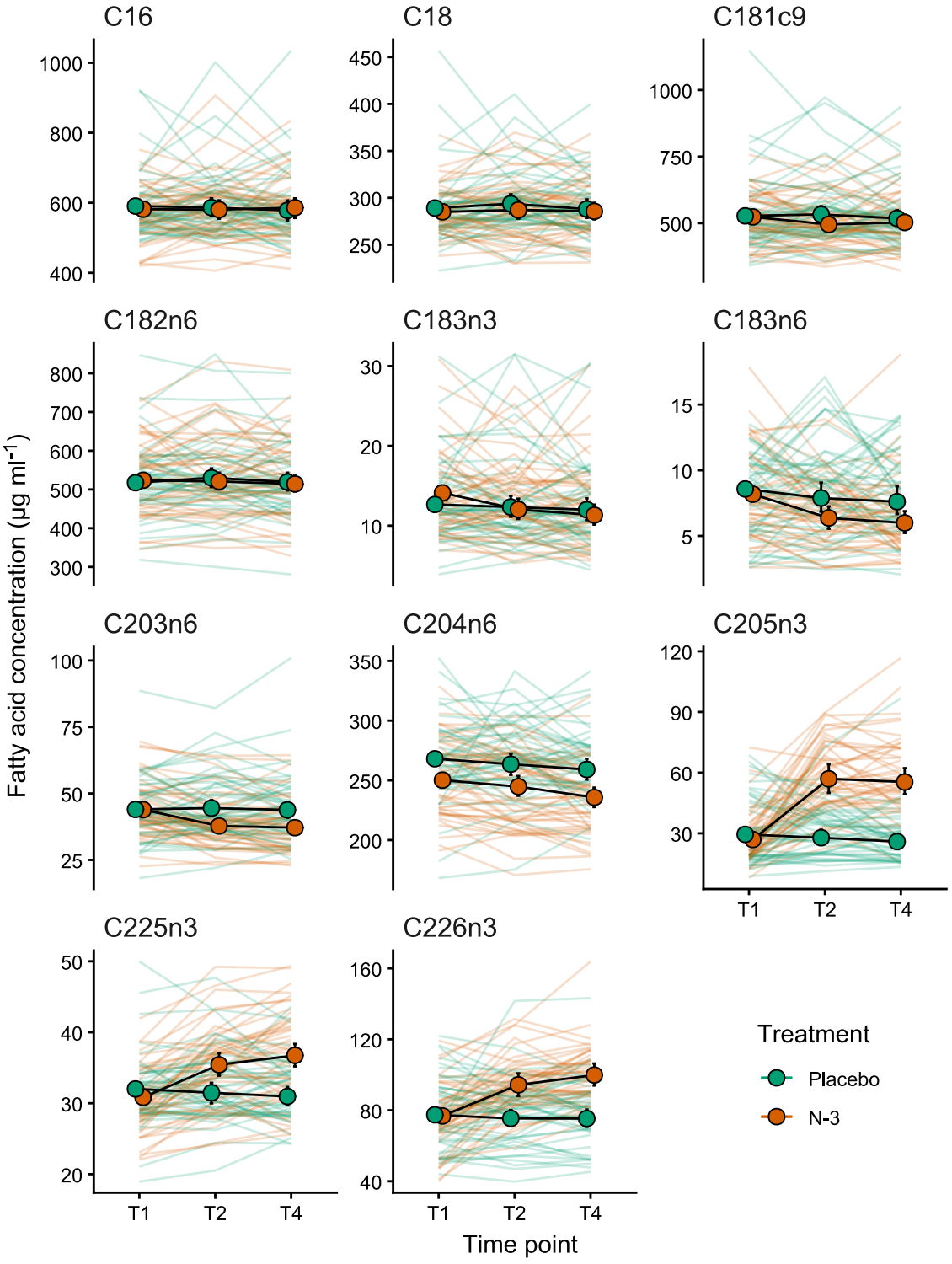


Figure 8: Predictions of fatty acid concentration in response to treatment. Pre-intervention values are means per treatment group, post-intervention means are predicted from the model with 99% PI.

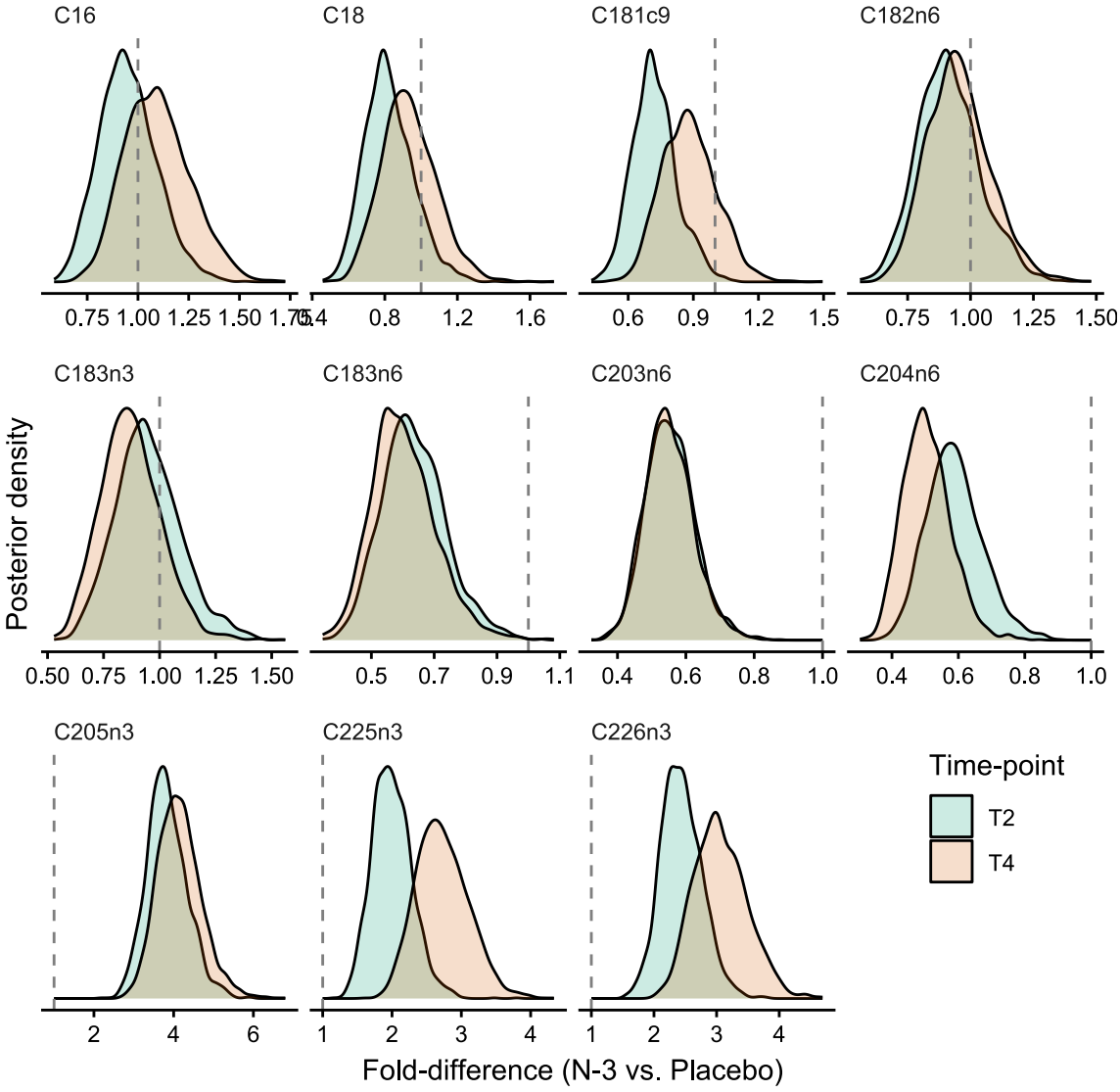


Figure 9: Posterior densities for differences between treatment groups in fatty acid concentrations.

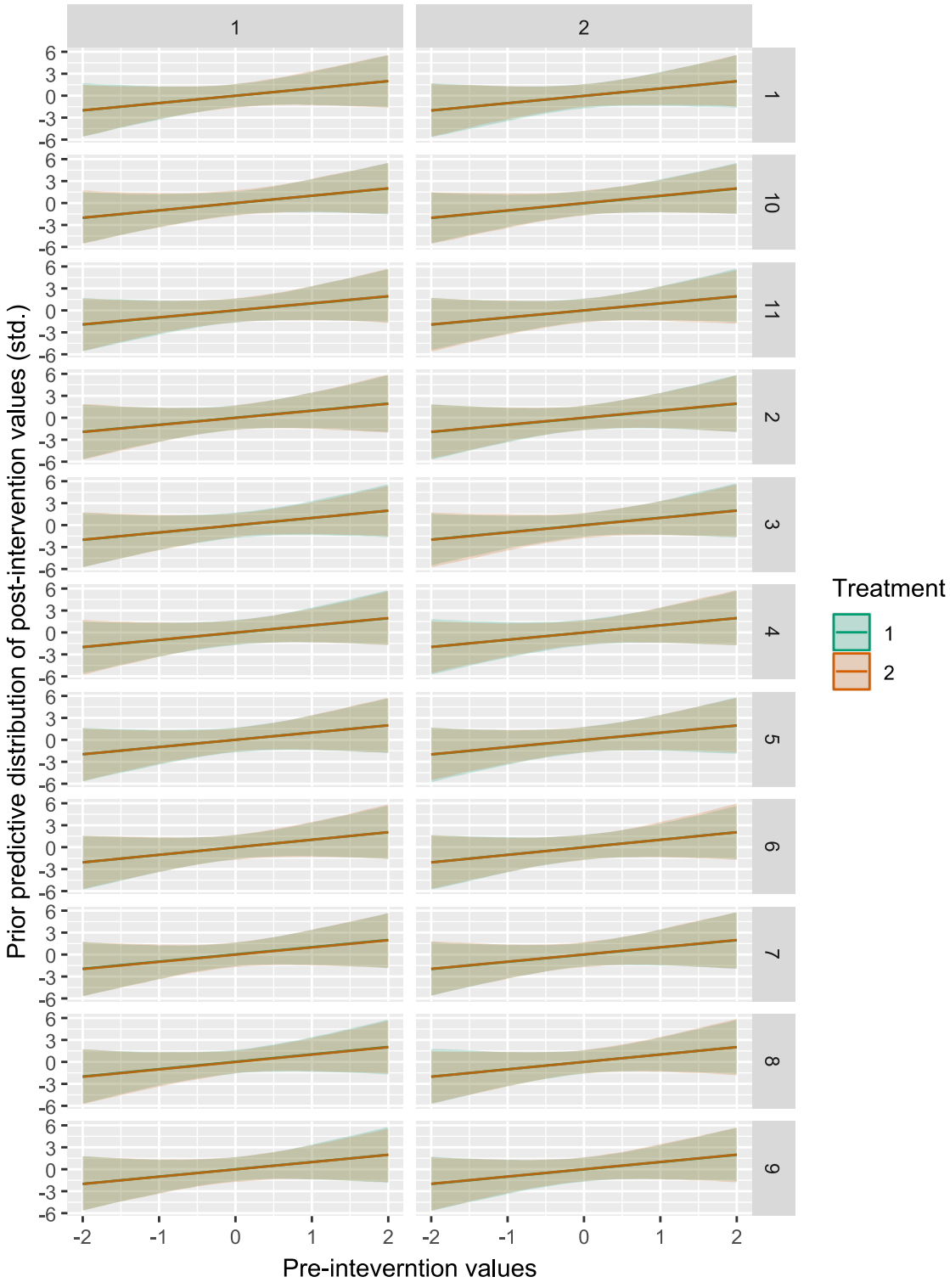


Figure 10: Prior predictive check model 3 (data imputation).

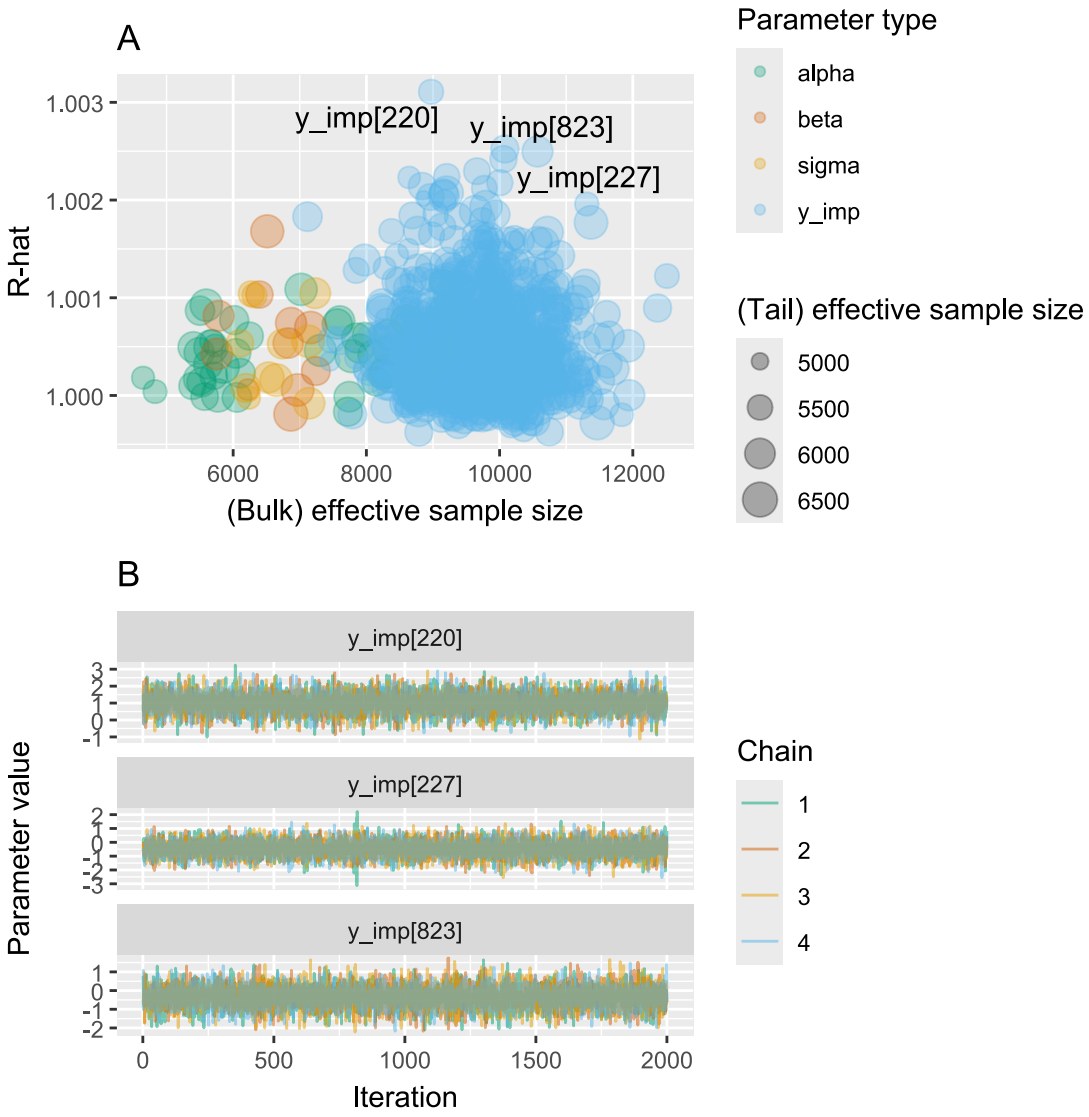


Figure 11: Model performance metrics model 3.

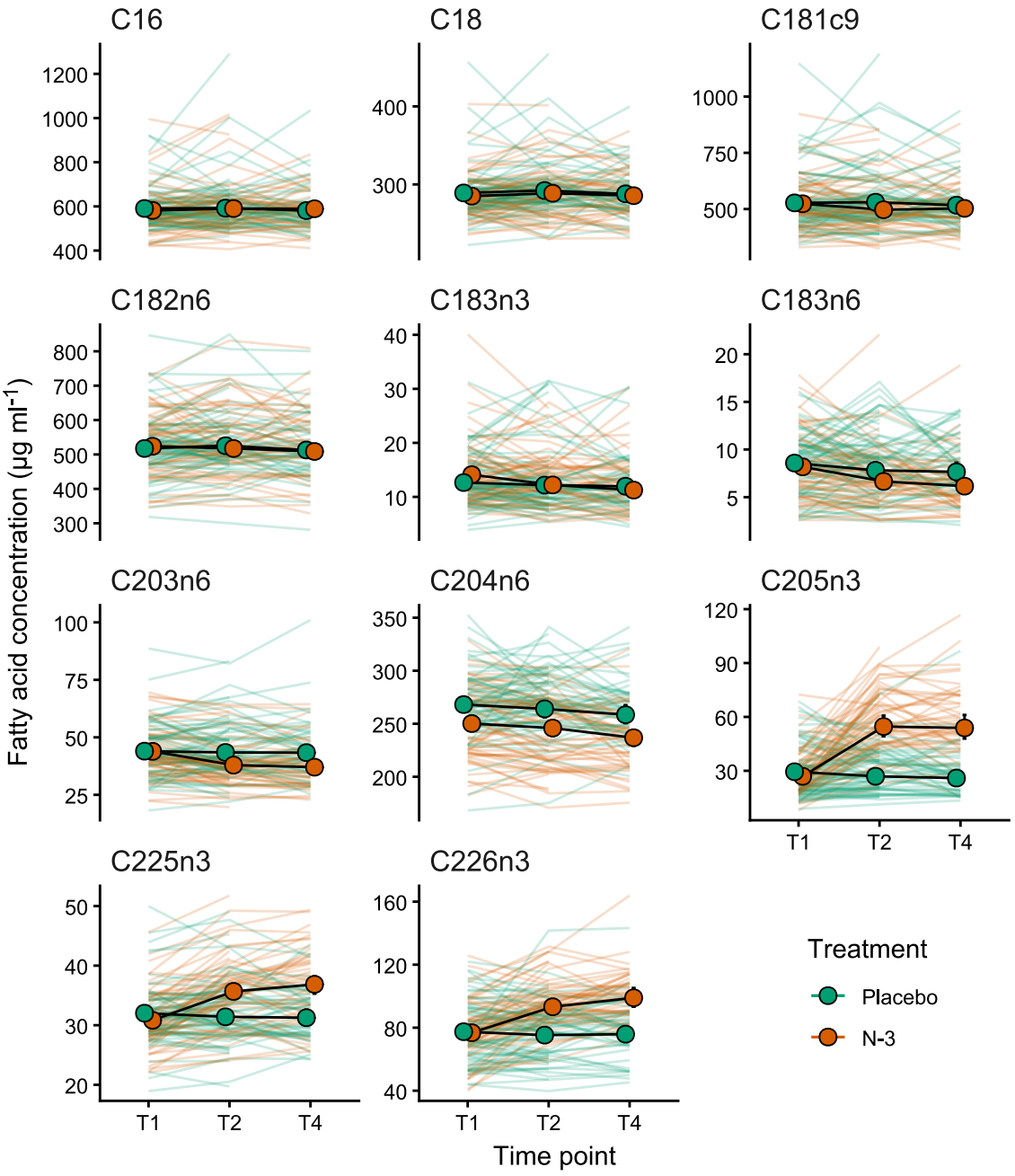


Figure 12: Model predictions. Pre-intervention values are observed means, post-treatment (T2, T4) values are model predictions conditional pre-intervention value with 99% percentile intervals.

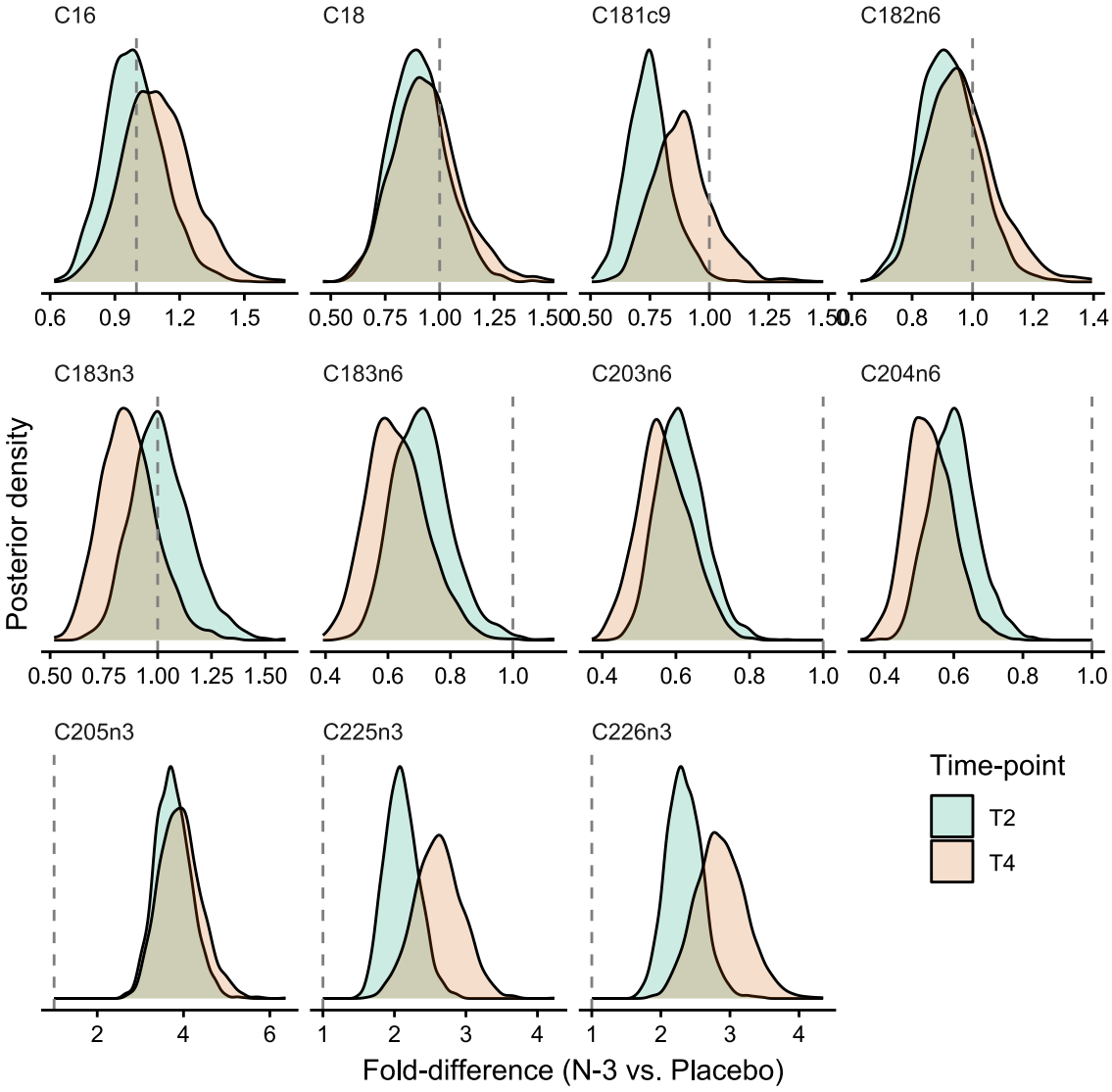


Figure 13: Posterior densities for differences between treatment groups in fatty acid concentrations (data imputation model).

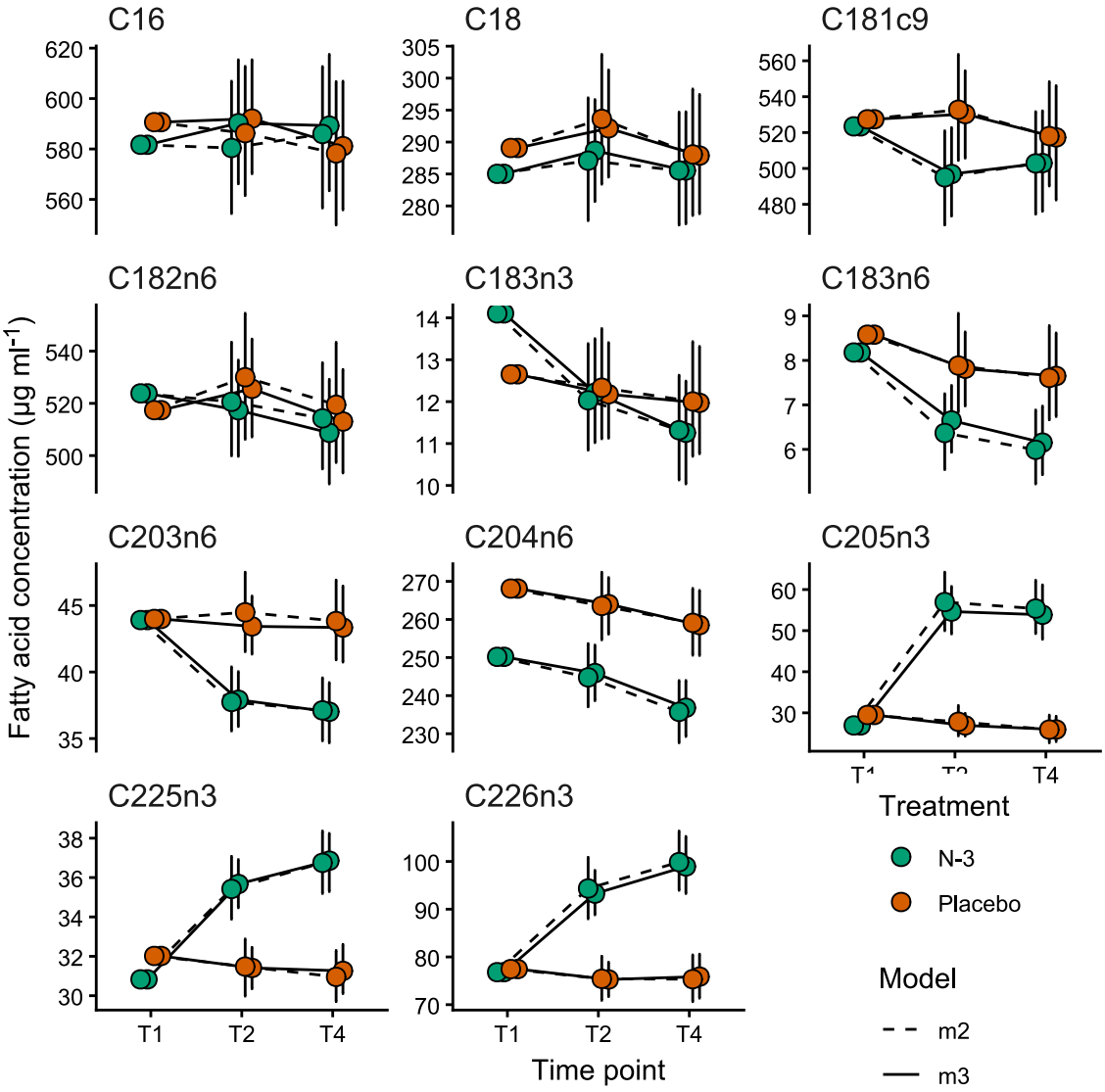


Figure 14: Posterior predictions for fatty acid concentrations in complete-case analysis (model m2) and data imputation model (model m3).

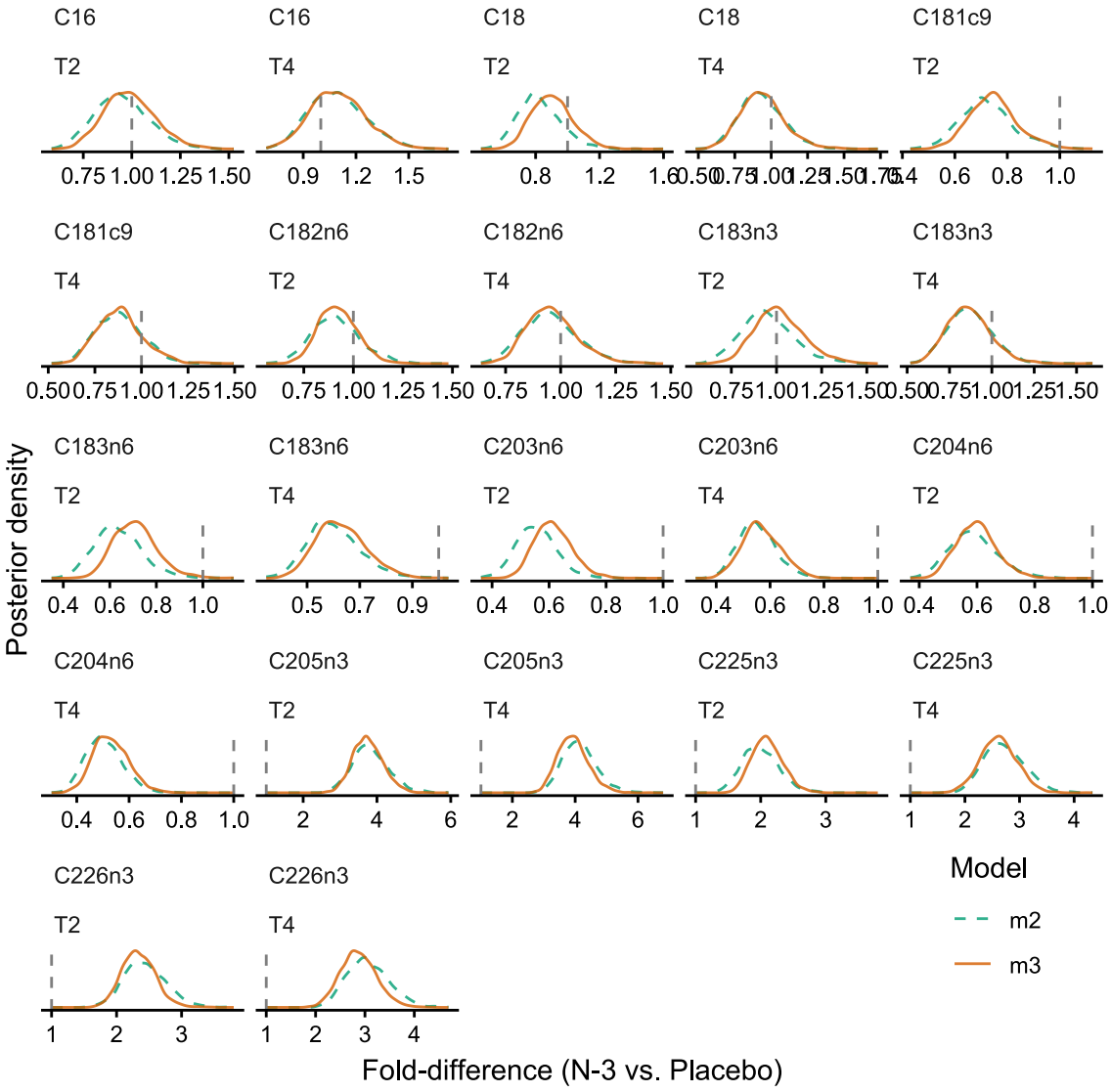


Figure 15: Comparing inference from complete-case and imputation models.

Does omega-3 supplementation, strength training, or their combination reduce the prevalence of metabolic syndrome compared to placebo?

A first analysis defines metabolic syndrome prevalence as the proportion of participants identified as having at least three out of five diagnostic criteria for metabolic syndrome. The analysis is complete-case analysis including individuals with at least three out of five criteria measurements over three studies time-point (pre-intervention, T1; post-supplementation, T2; post-supplementation + strength training, T4 Table 2).

Table 2

	Time		
	T1	T2	T4
	Male		
Placebo	21 (19%)	21 (19%)	21 (33%)
N-3	19 (26%)	19 (16%)	19 (26%)
	Female		
Placebo	16 (19%)	16 (12%)	16 (19%)
N-3	21 (24%)	21 (10%)	21 (19%)
	Total		
Placebo	37 (19%)	37 (16%)	37 (27%)
N-3	40 (25%)	40 (12%)	40 (22%)

Data are presnted as n (Metabolic syndrom proportion)

Table 3

The proportion of individuals identified as having metabolic syndrome (three out of five criteria thresholds) did not differ between Placebo and N-3 supplementation after the supplementation period or after the period with supplementation and strength training.

Intention-to-treat analysis with imputed missing values

Bayesian ITT-models

The goal here is to develop a Bayesian ITT model with missing data imputation to compare the treatment groups across all subcategories of the metabolic syndrome diagnostic criteria.

A first model is built to retrieve sub-score treatment effects (Blood pressure, HDL, Resting glucose, Triglycerides and Waist circumference).

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